CAPABILITY STATEMENT

Evidence-Bounded AI Support for the Federal Acquisition Lifecycle

CDC Reverse Industry Day | Notice 75D301-26-RFI-73483

Respondent	LumenCore	**Notice**	75D301-26-RFI-73483
Contact	Robert Ashworth	**Status**	Prototype-stage / in development
Location	Nashville, Tennessee	**Response**	AI acquisition support capability

Executive Summary

LumenCore is a founder-built, prototype-stage AI evidence and evaluation platform. It is designed to help acquisition and technical review teams determine what an AI-assisted workflow measured, which sources and assumptions it used, how it compared with a locked incumbent baseline, and which conclusions remain unsupported. The product is still in development and is not represented as federally deployed, commercially validated, certified, or authorized for autonomous acquisition decisions.

For CDC, LumenCore could be demonstrated as a bounded evidence layer around acquisition planning, market research, requirements development, proposal evaluation support, contract administration, and closeout. Human officials retain all decision and approval authority.

Acquisition Lifecycle Alignment

- Planning and market research: register the acquisition question, source inventory, update times, assumptions, and evidence gaps before analysis begins.
- Requirements development: trace each requirement to source material, acceptance criteria, exclusions, and reviewer ownership.
- AI tool and workflow evaluation: compare candidate workflows against a preselected manual or non-AI baseline using locked metrics and holdout rules.
- Proposal evaluation support: preserve evidence citations, scoring rationale, uncertainty, conflicts, abstentions, and human reviewer decisions without automating source selection.
- Administration and closeout: assemble milestone, deliverable, exception, and disposition records into machine-readable manifests and concise reviewer packets.

Technical Approach

1. Source gate - record origin, owner, timestamp, access boundary, hash, transformation, and measured-versus-synthetic status.
2. Evaluation-plan gate - freeze the baseline, candidate, metric, threshold, holdout policy, and prohibited claims before scoring.
3. Replay gate - execute the approved comparison, preserve configuration and failures, and block results when source quality or required controls fail.
4. Review gate - generate a machine-readable manifest plus a human-readable proof card showing improvements, regressions, uncertainty, missing evidence, and claim boundaries.
5. Decision gate - require an authorized human to approve any recommendation, pilot transition, external transmission, or high-impact action.

Current Product and Evidence Status

- Working prototype components include source inventory, provenance records, locked evaluation plans, baseline-versus-candidate replay, evidence manifests, reviewer summaries, and fail-closed workflow controls.
- Public reviewer surface: https://lumen-core.ai/proof_to_pilot.html
- Evidence is internally reproducible and hash-addressed; it is not represented as CDC validation, field performance, realized savings, ATO, FedRAMP authorization, or CMMC certification.
- LumenCore is available for a controlled, non-production demonstration using synthetic, public, or CDC-authorized data and pre-agreed evaluation criteria.

Proposed Reverse Industry Day Demonstration

LumenCore would demonstrate a small acquisition workflow in which a requirement and evaluation plan are registered before scoring; two AI-assisted approaches are compared with a locked baseline; provenance, configuration, exceptions, and uncertainty are captured; and a reviewer packet distinguishes measured evidence from assumptions. The demonstration would emphasize auditability, abstention when evidence is insufficient, and human control rather than autonomous acquisition authority.

Contact

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